

Zhengkao (Daniel) Yu

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EDUCATION

The University of Hong Kong

BEng in Computer Engineering

- GPA: **3.40/4.3 (Top 15%, 2558 students)**
- *Relevant Coursework: Probabilistic Systems Analysis, Data Structures and Algorithms, Operating Systems, Embedded Systems, Digital System Design, Pattern Recognition and Machine Intelligence, etc.*

Hong Kong SAR
Sept 2023–Jun 2027

Peking University

Academic Exchange Program

- *Relevant Coursework: Simulation Methods for Optimization and Learning (A)*

Beijing, China
Jun 2024–Aug 2024

PUBLICATION

- **Zhengkao Yu***. "Design and Wizard-of-Oz Evaluation of a Tri-Modal Contactless In-Vehicle HMI for High-Noise and High-Vibration Cabins." *Journal of Intelligence and Knowledge Engineering (JIKE)*, Vol. 4, No. 2, April 2026. DOI: [10.62517/jike.202604230](https://doi.org/10.62517/jike.202604230). (Published)
- **Zhengkao Yu***. "Game-Diffusion-based Multi-Agent Learning of Ramp Merging Coordination for Connected and Automated Vehicles." *IEEE OJITS*, 2026 (Accepted/ In Press).
- **Zhengkao Yu***. "Design and Implementation of a Wheeled Robot with Obstacle Avoidance and Navigation Capabilities for Drug Delivery." *ICEEGT 2025*, pp. 704-720. (Published)

RESEARCH EXPERIENCE

Adversarial Testing for Autonomous Driving

Research Assistant, supervised by Prof. Hao Zhang (CMU & UTA), Remote

Jun 2025–Present

- Formulated the discovery of systematic failures in highway-merging planning systems as a non-cooperative game-theoretic problem, addressing the inherent limitation of random or hand-crafted testing in exposing edge-case vulnerabilities
- Proposed and developed a novel multi-agent reinforcement learning (MARE) framework integrated with diffusion-based generative models to synthesize realistic and causally coherent adversarial traffic behaviors that challenge the planning module
- Generated 2000+ valid safety-critical scenarios in HighwayEnv (PyTorch implementation); achieved 92% effectiveness in triggering unexpected planning behaviors
- Designed and implemented a SOTIF-aligned evaluation protocol; improved failure detection rate by 85% compared to random testing baselines, and reduced the collision risk of the tested planning module from 12.3% to 3.7% through targeted adversarial validation
- Authored a first-author paper based on this work, accepted by the IEEE Open Journal of Intelligent Transportation Systems (OJITS), 2026, which formalizes the game-diffusion framework for multi-agent ramp merging coordination

Human – Computer Interaction for Extreme Environments

Independent Researcher, supervised by Prof. Xiang Chen (UCLA), Remote

Jan 2026–Present

- Addressed the reliability degradation of conventional in-vehicle interfaces (touch/voice) in high-noise and high-vibration environments by formulating the interaction as a multimodal signal fusion and decision-confirmation problem
- Proposed a lightweight multimodal interaction paradigm integrating gaze selection, head confirmation, and noise-robust speech refinement
- Designed a three-layer perception – understanding – feedback architecture for multimodal fusion and conflict resolution

- Developed an interactive prototype and implemented a Wizard-of-Oz evaluation framework with 30+ operator participants, quantifying performance via NASA-TLX cognitive load metrics and task completion efficiency
- Authored a first-author publication based on this work in the Journal of Intelligence and Knowledge Engineering (JIKE), April 2026, which details the system architecture and experimental validation

Simulation, Adaptive Control and Optimization Framework for Reliable Robotic Manipulation

May 2026 – Present

Independent Researcher, supervised by Dr. Liangcai Tan (Archimedes Controls Corp.), Remote

- Developed a nonlinear 2-DOF robotic manipulator framework in MATLAB/Simulink with torque-aware PID and Fuzzy-PID control, trajectory retiming, inverse-dynamics feedforward, and actuator constraints
- Built a Bayesian optimization and robustness-testing workflow covering payload variation, model uncertainty, disturbances, sensor noise, saturation, and robot-configuration changes.
- Evaluated four fixed controllers across five prespecified cases and three simulation time profiles, completing 60/60 runs and 40/40 paired comparisons. All 20 plant-only refinements remained within tolerance; 14/20 whole-loop refinements were stable, while six revealed feedback-sampling sensitivity.
- Preparing a first-author manuscript on the simulation framework, control methods, and quantitative validation.

Robust Autonomous Navigation for Wheel-Leg Hybrid Robot

Apr 2025–Jul 2025

Independent Researcher, supervised by Prof. Naira Hovakimyan (UIUC), Remote

- Designed and validated a hybrid wheel-leg robotic platform for autonomous drug delivery in constrained hospital environments
- Developed an EKF-based multi-sensor fusion framework integrating LiDAR, IMU, ultrasonic, and infrared sensors for robust localization under dynamic obstacles
- Proposed an integrated motion planning architecture combining JPS-optimized A* global planning with DWA local obstacle avoidance for real-time adaptation
- Implemented PID-based self-balancing control (tilt error $\pm 2^\circ$) and multi-mode locomotion
- Validated through real-world hospital experiments: 98.2% obstacle avoidance success rate, ≤ 5 cm localization error, ≤ 10 cm path tracking error, 0.3 s dynamic obstacle response
- Authored a first-author conference paper based on this work at ICEEGT 2025, which details the integrated navigation architecture and experimental findings

INDUSTRY EXPERIENCE

SevnceRobotics

Shanghai, China

Electrical Engineering Intern, Product R&D Center

Jun 2025 – Aug 2025

- Designed dual-layer CAN architecture (CAN FD/CAN C) and 24-series lithium battery BMS for humanoid robotic platforms
- Reduced communication latency by 12% and improved anti-interference robustness in industrial deployment scenarios
- Leveraged Altium Designer to complete schematic design and 4-layer PCB layout for sensor acquisition board; achieved 99.9% data transmission success rate after physical debugging; used MATLAB for system simulation and reliability testing
- Cross-collaborated with Control Algorithm, Product Testing and Procurement teams to align technical requirements, formulate test cases and output preliminary BOM; participated in HR160 electrical architecture V0.3 design reviews and proposed adopted optimization improvements

Ragtime Lab, Zhenjiang Pioneer Cropscience Co., Ltd.

Zhenjiang, China

Robotics Research Intern

July 2026 – Sept 2026

- Curated and standardized a multimodal dataset for 33 industrial manipulation tasks spanning packaging,

production, material handling, laboratory operations and general manipulation; integrated task documentation and visual records with synchronized dual-arm poses, joint states, gripper signals and timestamps for feasibility analysis and benchmarking

- Developed a task-driven robot design and installation optimization framework in Python, PyTorch and MuJoCo for 13 robotic arms, integrating URDF-based kinematics, CUDA-accelerated NSGA-II/Pareto search, continuous SE(3) inverse kinematics, collision-aware branch selection, robustness analysis and joint-velocity retiming
- Designed a fixed-time mount evaluation pipeline for PiperX dual-arm manipulation across 27 trajectories from 12 task families and four mounting configurations (108 experiment cells), preserving source timestamps while enforcing joint limits, self/cross-arm/table/swept-transition collision checks and mount-topology constraints
- Built a reproducible evaluation and reporting workflow with per-frame result shards, provenance hashes, automated tests, aggregate metrics and synchronized MuJoCo videos; reproduced and validated 102 multi-robot trials covering 49,302 trajectory points and produced four-arm, two-arm and 13-arm comparison reports

ADDITIONAL INFORMATION

Technical Skills:

- Programming: Python (PyTorch/OpenCV), C/C++ (Embedded/ROS), MATLAB/Simulink, Shell Script, Verilog/SystemVerilog, SQL
- Tools: Altium Designer, CANoe, Git/GitHub, VS Code, LaTeX
- ML: PyTorch (MARL/Diffusion), ROS 2, EKF/UKF, PID, A*/JPS/DWA, OpenCV

Interests: Robotics Architecture, Robotics System Design, Autonomous Driving (Planning & Testing), Multimodal HCI, Embedded Systems

Languages: Chinese (Native), English (Fluent, IELTS 7)